# DESIGN OF RRM WALL 4m Height Akhnoor Poonch Package 7

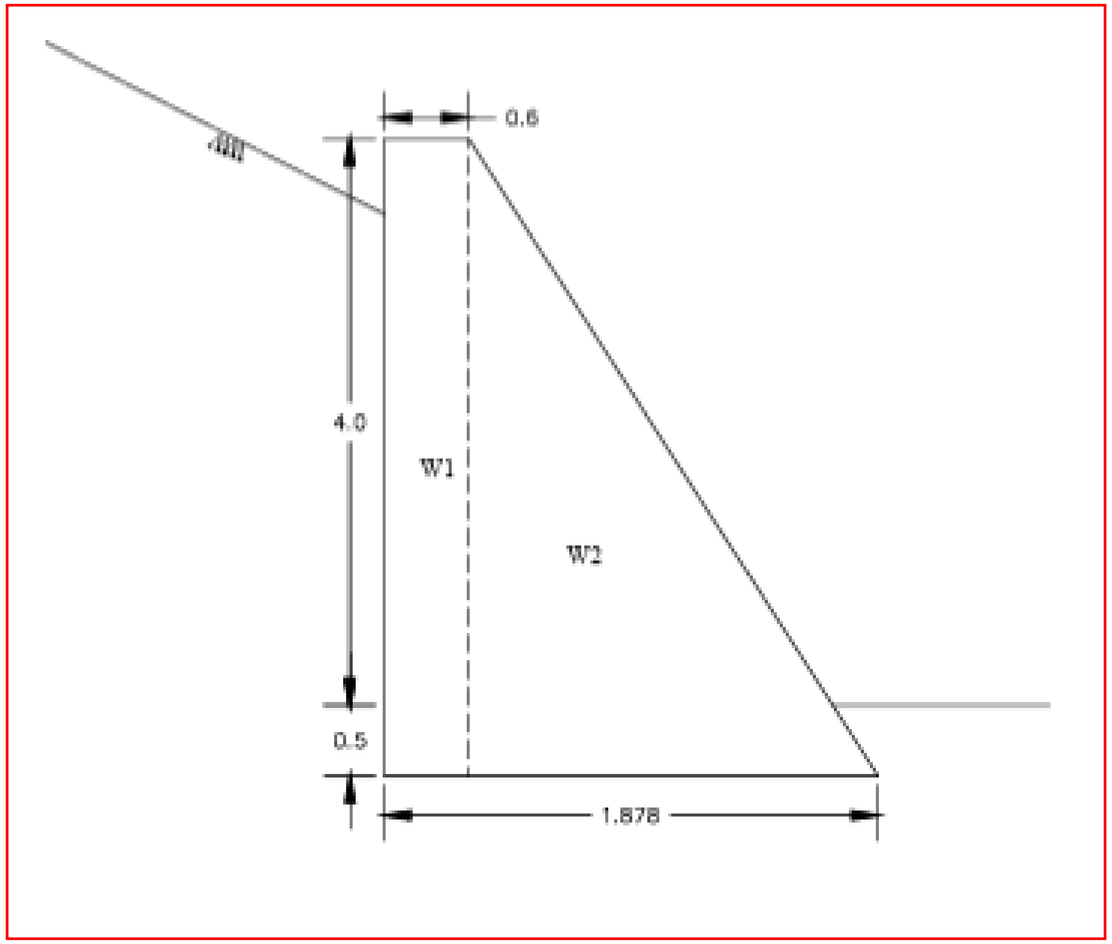
| INPUT DATA | | | |
|---|---|---|---|
| Unit wt. of RRM stone | $\gamma_C$ | 23.50 | $kN/m^3$ |
| Unit wt. of water | $\gamma_W$ | 9.81 | $kN/m^3$ |
| Unit wt. of soil(Saturated) | $\gamma_{Sat}$ | 21.0 | $kN/m^3$ |
| Angle of Backfill Slope | i | 20.00 | degrees |
| Angle of wall with Vertical | $\alpha$ | 0.00 | degrees |
| Angle of Friction between wall & earth fill | $\delta=2/3*\phi$ | 21.33 | degrees |
| Length of Foundation Base | L | 1.60 | m |
| Total height | H | 4.00 | m |
| Horizontal co-eff. of acceleration, | $\alpha_h=\alpha_o*I*\beta$ | 0.150 | g |
| Vertical co-eff. of acceleration, | $\alpha_v=\alpha_h*2/3$ | 0.100 | g |
| Angle of internal Friction of soil | $\phi$ | 32.0 | deg. |

| SLIDING SURFACE PARAMETERS | | | |
|---|---|---|---|
| Cohesion, c | | 15.0 | kPa |
| Angle of friction, f | | 32.0 | deg |
| Factor Safety for Sliding (As per IS 14458 Part-2) | | | |
| Normal Case | | 1.5 | |
| Earthquake Case | | 1.2 | |
| Factor Safety for Over Turning (As per IS 14458 Part-2) | | | |
| | Normal Case | 2.0 | |
| | Earthquake Case | 1.5 | |

| Earth Pressure Co-efficients | | | |
|---|---|---|---|
| Co-efficient of Active Earth Pressure | $C_{a,\ Normal}$ | 0.374 | |
| Co-efficient of Passive Earth Pressure | $C_{p,\ Normal}$ | 2.670 | |
| Co-efficient of Active Earth Pressure | $C_{a,\ Earthquake}$ | 0.620 | |
| Co-efficient of Passive Earth Pressure | $C_{p,\ Earthquake}$ | 1.613 | |

**Self Weight Calculations:**

| Wall | Wt. | ex | ey | Wt.*ex | Wt.*ey |
|---|---|---|---|---|---|
| 1 | 50.00 | 1.35 | 2.00 | 67.50 | 100.00 |
| 2 | 46.75 | 0.73 | 1.13 | 34.28 | 52.98 |
| | **96.75** | **1.05** | **1.58** | **101.78** | **152.98** |



# DESIGN OF RRM wall 4.0m Height Akhnoor Poonch Package 7
## EARTH PRESSURE COEFFICIENT CALCULATIONS

**INPUT:**

| | | | | |
|---|---|---|---|---|
| Unit weight of saturated soil | $\gamma_s$ | = | 21.00 | $kN/m^3$ |
| Angle of internal Friction of soil | $\phi$ | = | 32.00 | deg. |
| Inclination of Back Fill | $i$ | = | 20.00 | deg. |
| Unit weight of RRM stone | $\gamma_c$ | = | 23.50 | $kN/m^3$ |
| Unit weight of water | $\gamma_w$ | = | 9.81 | $kN/m^3$ |

**Active Earth Pressure Coefficients: for Vertical wall face with back fill**

*(seismic coefficient method, IS 1893-2014 Part 3 & IS 1893-2016 Part 1)*

| | | | | | |
|---|---|---|---|---|---|
| Horizontal coefficient of acceleration, | $\alpha_h=\alpha_o$*I*$\beta$ | $\alpha_h$ | = | 0.000 | g |
| Vertical Coefficient of acceleration | $\alpha_v=\alpha_h$*2/3 | $\alpha_v$ | = | 0.000 | g |
| Angle of earth fill face with vertical | | $\alpha$ | = | 0.000 | deg. |
| Angle of Friction between wall & earth fill | $\delta$=2/3*$\phi$ | $\delta$ | = | 21.33 | deg. |
| | | $\lambda(\pm)$ | = | $\tan^{-1}\dfrac{\alpha_h}{1\pm\alpha_v}$ | |
| | | $\lambda(\pm)$ | = | 0.00 | deg. |
| | | 0.6 | = | 0.00 | deg. |

Coefficient of active earth pressure, $C_a(\pm)$

(cl.8.1.1, IS 1893-1984, pg.46)

$$C_a(\pm) = \frac{(1\pm\alpha_v)\cos^2(\phi-\lambda-\alpha)}{\cos\lambda\cos^2\alpha\cos(\partial+\alpha+\lambda)} * \left[\frac{1}{1+\left\{\frac{\sin(\theta+\partial)\sin(\phi-i-\lambda)}{\cos(\alpha-i)\cos(\partial+\alpha+\lambda)}\right\}^{\frac{1}{2}}}\right]^2$$

| | | | | |
|---|---|---|---|---|
| | = | 0.772 | 0.485 | |
| $C_a(+)$ | | **0.374** | -------(1) | |
| | = | 0.772 | 0.485 | |
| $C_a(-)$ | | **0.374** | -------(2) | |
| Taking maximum of 1 & 2; $C_{a(Normal)}$ | = | **0.374** | | |

**Passive Earth Pressure Coefficients : for Vertical wall face with back fill**

$C_{p(Normal)}$ = $1/C_{a(Normal)}$ $C_{p(Normal)}$ = 2.670

**Active Earth Pressure Coefficients with earthquake: for Vertical wall face with back fill**

*(seismic coefficient method, IS1893-1984)*

| | | | | | |
|---|---|---|---|---|---|
| Horizontal coefficient of acceleration, | $\alpha_h=\alpha_o$*I*$\beta$ | $\alpha_h$ | = | 0.150 | g |
| Vertical Coefficient of acceleration | $\alpha_v=\alpha_h$*2/3 | $\alpha_v$ | = | 0.100 | g |
| Angle of earth fill face with vertical | | $\alpha$ | = | 0.000 | deg. |
| Angle of Friction between wall & earth fill | $\delta$=2/3*$\phi$ | $\delta$ | = | 21.33 | deg. |
| | | $\lambda(\pm)$ | = | $\tan^{-1}\dfrac{\alpha_h}{1\pm\alpha_v}$ | |
| | | $\lambda(+)$ | = | 7.77 | deg. |
| | | $\lambda(-)$ | = | 9.46 | deg. |

$$C_a(\pm) = \frac{(1\pm\alpha_v)\cos^2(\phi-\lambda-\alpha)}{\cos\lambda\cos^2\alpha\cos(\partial+\alpha+\lambda)} * \left[\frac{1}{1+\left\{\frac{\sin(\theta+\partial)\sin(\phi-i-\lambda)}{\cos(\alpha-i)\cos(\partial+\alpha+\lambda)}\right\}^{\frac{1}{2}}}\right]^2$$

| | | | | |
|---|---|---|---|---|
| | = | 0.960 | 0.622 | |
| $C_a(+)$ | | **0.597** | -------(3) | |
| | = | 0.906 | 0.684 | |
| $C_a(-)$ | | **0.620** | -------(4) | |
| Taking maximum of 3 & 4; $C_{a(Earthquake)}$ | = | **0.620** | | |

**Passive Earth Pressure Coefficients with earthquake: for Inclined wall face with back fill**

$C_{p(Earthquake)}$ = $1/C_{a(Earthquake)}$ $C_{p(Earthquake)}$ = 1.613

| DESIGN OF RRM wall 4.0m Height for Akhnoor Poonch Package 7 | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **S. No.** | **Load Case Description** | **Loads** | **Vertical Force** | **Lever arm** | **Horizontal Force** | **Lever arm** | **Resisting Moments** | **Over turning Moments** | **Sum of Resisting moments** | **Sum of overturning moments** | **F.O.S. Calculated** | **F.O.S. Allowable** |
| | | | **V** | **LX** | **P** | **LY** | **MR** | **MO** | **∑MR** | **∑MO** | **MR / MO** | **As per IS 14458 (Part-2)** |
| | | | **kN/m** | **m** | **kN/m** | **m** | **kNm/m** | **kNm/m** | **kNm/m** | **kNm/m** | | |
| 1 | Case-1 (Normal Case) | Self weight | 96.75 | 1.05 | | | 101.78 | | 168 | 63 | 2.67 | 2.00 |
| | | Active soil pressure | 41.55 | 1.60 | 47.24 | 1.33 | 66.48 | 62.98 | | | | |
| 2 | Case-2 (Normal Case with DBE) | Self weight | 96.75 | 1.05 | | | 101.78 | | 212 | 127 | 1.67 | 1.50 |
| | | Earthquake load (30% Vertical & Full Horizontal) | 2.90 | 1.05 | 14.51 | 1.33 | | 22.40 | | | | |
| | | Active soil pressure | 41.55 | 1.60 | 47.24 | 1.33 | 66.48 | 62.98 | | | | |
| | | Active Soil pressure (dynamic increment) | 27.22 | 1.60 | 30.95 | 1.33 | 43.55 | 41.26 | | | | |

# DESIGN OF RRM wall 4.0m Height for Akhnoor Poonch Package 7

<table>
<tr><th rowspan="3">S. No.</th><th rowspan="3">Load Case Description</th><th rowspan="3">Loads</th><th>Vertical Force</th><th>Horizontal Force</th><th>Sum of Vertical Forces</th><th>Sum of Horizontal Forces</th><th>F.O.S. Calculated</th><th rowspan="3">F.O.S. Allowable (As per IS 14458 Part-2)</th></tr>
<tr><th>V</th><th>P</th><th>∑V</th><th>∑P</th><th rowspan="2">$F = \frac{(\Sigma V)\tan\Phi + F_p}{\Sigma P}$</th></tr>
<tr><th>kN/m</th><th>kN/m</th><th>kN/m</th><th>kN/m</th></tr>
<tr><td rowspan="3">1</td><td rowspan="3">Case-1<br>(Normal Case)</td><td>Self weight</td><td>96.75</td><td></td><td rowspan="3">138</td><td rowspan="3">47</td><td rowspan="3">2.04</td><td rowspan="3">>1.5</td></tr>
<tr><td>Soil weight & Soil pressure</td><td>41.55</td><td>47.24</td></tr>
<tr><td>Passive pressure</td><td></td><td>10.09</td></tr>
<tr><td rowspan="5">2</td><td rowspan="5">Case-2<br>(Normal Case with DBE)</td><td>Self weight</td><td>96.75</td><td></td><td rowspan="5">168</td><td rowspan="5">93</td><td rowspan="5">1.20</td><td rowspan="5">>1.2</td></tr>
<tr><td>Earthquake load (30% Vertical & Full Horizontal)</td><td>2.90</td><td>14.51</td></tr>
<tr><td>Soil weight & Soil pressure</td><td>41.55</td><td>47.24</td></tr>
<tr><td>Soil pressure (dynamic increment)</td><td>27.22</td><td>30.95</td></tr>
<tr><td>Passive pressure</td><td></td><td>6.10</td></tr>
</table>

## DESIGN OF RRM wall 4.0m Height for Akhnoor Poonch Package 7

| S. No. | Load Case Description | Loads | Vertical Force | Lever arm | Horizontal Force | Lever arm | Moments | Sum of Moments | Sum of Vert. For. | Eccentricity (m) | Stresses at Ends ($kN/m^2$) | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | V | LX | P | LY | M | ∑M | ∑V | e=L/2-SM/SV | s = SV/L*(1±6e/L) | |
| | | | kN/m | m | kN/m | m | kNm/m | kNm/m | kNm/m | | sA | sB |
| 1 | Case-1 (Normal Case) | Self weight | 96.75 | 1.05 | | | 101.78 | 105.28 | 138.30 | 0.039 | 99.00 | 73.87 |
| | | Soil weight & Soil pressure | 41.55 | 1.60 | 47.24 | 1.33 | 3.49 | | | | | |
| 2 | Case-2 (Normal Case with DBE) | Self weight | 96.75 | 1.05 | | | 101.78 | 91.27 | 168.42 | 0.26 | 207.14 | 3.38 |
| | | Earthquake load (30% Vertical & Full Horizontal) | 2.90 | 1.05 | 14.51 | 1.33 | -16.30 | | | | | |
| | | Soil weight & Soil pressure | 41.55 | 1.60 | 47.24 | 1.33 | 3.49 | | | | | |
| | | Active Soil pressure (dynamic increment) | 27.22 | 1.60 | 30.95 | 1.33 | 2.29 | | | | | |

**PROJECT: BEARING CAPACITY CALCULATION FOR 4.0m HT RRM wall for Akhnoor Poonch Package 7**

## Safe Bearing Capacity Calculations

As per IS 6403-1981

| | | | | |
|---|---|---|---|---|
| Maximum Design Section | | = | 4 | m |
| Load consider | DL | = | 0 | $kN/m^2$ |
| | LL | = | 24 | $kN/m^2$ |
| Block width | B | = | 1.9 | m |
| Eccentricity | e | = | 0.039 | m. (From Design sheet) |
| Effective width | B-2e | = | 1.822 | |
| Water Table depth from G. L. | | = | 20 | m. |
| Water Table correction factor W' | | = | 1.00 | |
| Pressure at base of wall= | | | 99 | $kN/m^2$ (From Design sheet) |
| Required $q_{safe}$ | | = | 9.9 | t/m2 |

Following are the shear parameter of the soil of the top layer

| | | | |
|---|---|---|---|
| C | = | 15 | $kN/m^2$ |
| $\phi$ | = | 32 | Degree |
| $\gamma$ | = | 21 | kN/m3 |
| $D_f$ | = | 0.6 | m |

For $\phi$ = 32 Degree

| | | |
|---|---|---|
| $N_c$ | = | 35.49 |
| $N_q$ | = | 23.2 |
| $N_\gamma$ | = | 30.2 |

**Calculation of SBC**

Depth Factor

$dc = 1 + 0.2(D_f/B)(N\phi)^{1/2}$ = 1.1422

$dq = d\gamma = 1$ for $Ø < 10^0$

$d_q = d_\gamma = 1+0.1(D_f/B)(N\phi)^{1/2}$ for $Ø > 10^0$ = 1.0711

Shape Factor

$Sc = Sq = S\gamma =$ 1 (For Strip Footing)

Inclination Factor

$i_c = i_q = i_\gamma =$ 1 (For vertical Wall)

$q_{Ult} = C N_c S_c d_c i_c + q N_q S_q d_q i_q + 0.5 \gamma B' N_\gamma S_\gamma d_\gamma i_\gamma W'$ **(As per IS 6403:1981)**

= 738.07 $kN/m^2$

$q_{safe}$ = 295.23 $kN/m^3$ ( FOS=2.5)

29.52 t/m2 ***Safe***

**Depth of Foundation from Rankine's formula**

$$D_f = \frac{q_u}{\gamma}\left(\frac{1-\sin\varphi}{1+\sin\varphi}\right)^2$$

Df = 0.45 m ***Safe***